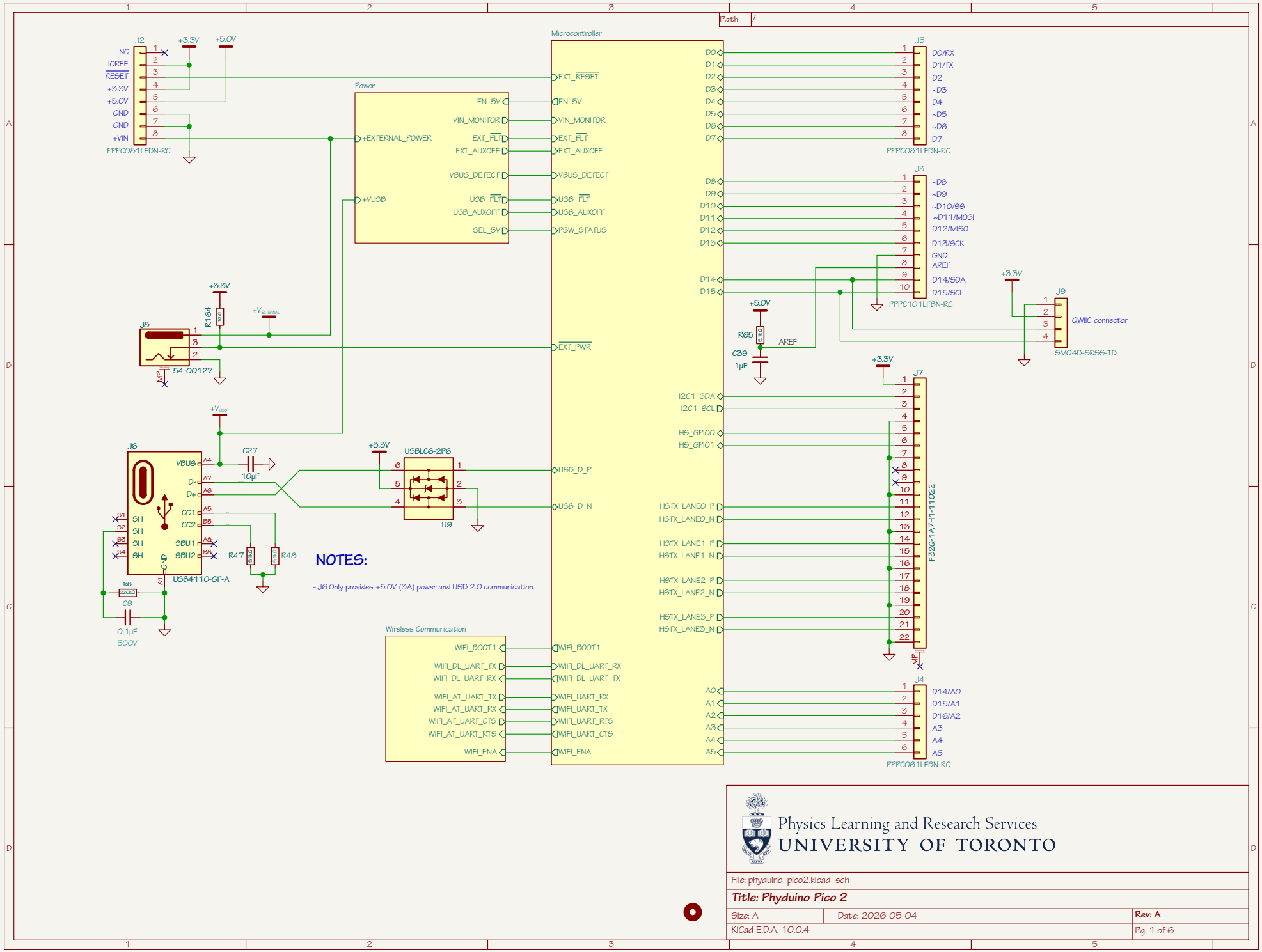
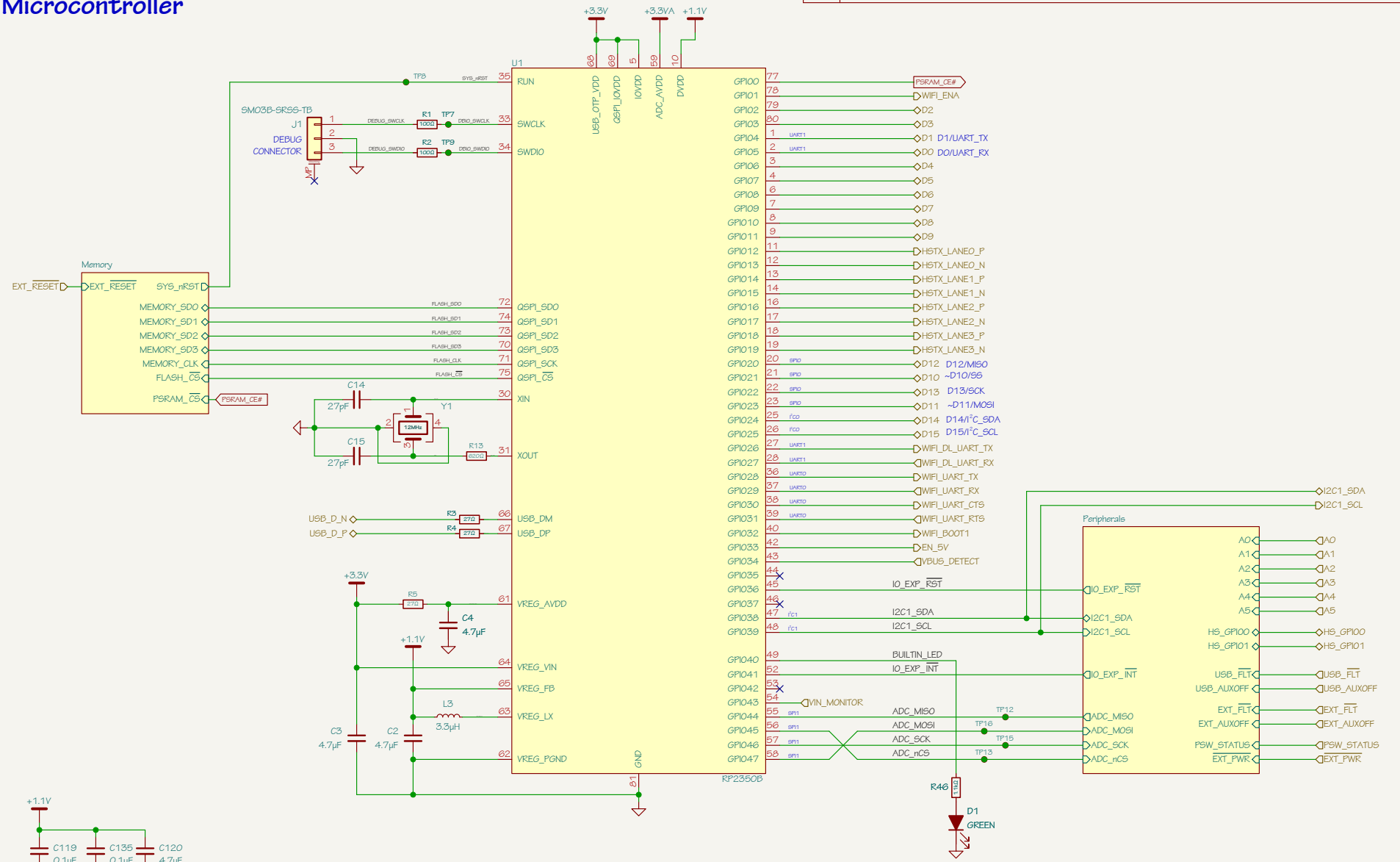

Path /
Microcontroller
Power
Wireless Communication
J2
PPPC081LFBN-RC
J5
PPPC081LFBN-RC
J3
PPPC101LFBN-RC
J9
QWIIC connector
SM04B-SRSS-TB
J7
F32Q-1A7H1-11022
J4
PPPC061LFBN-RC
J8
54-00127
J6
USB4110-GF-A
U9
USBLC6-2P6
R65
C39
1µF
C27
10µF
R47
R48
R6
C9
0.1µF
500V
NOTES:
- J6 Only provides +5.0V (3A) power and USB 2.0 communication.
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File: phyduino_pico2.kicad_sch
Title: Phyduino Pico 2
Size: A
Date: 2026-05-04
Rev: A
KiCad E.D.A. 10.0.4
Pg: 1 of 6

Microcontroller

Path /Microcontroller/



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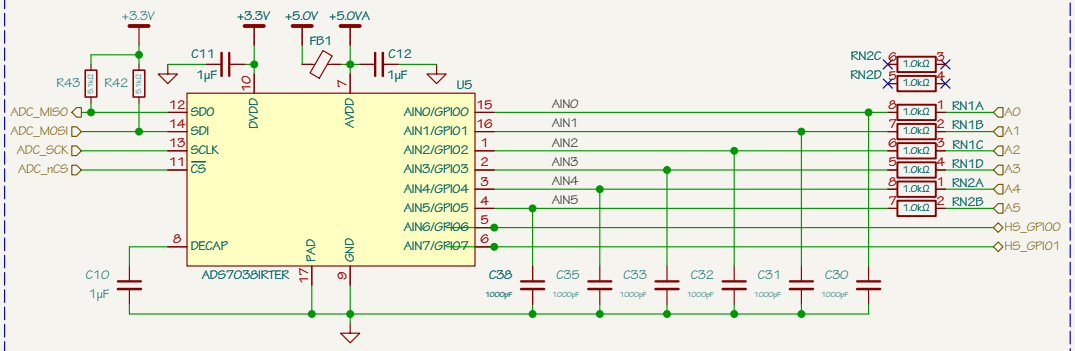
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Title: Phyduino Pico 2

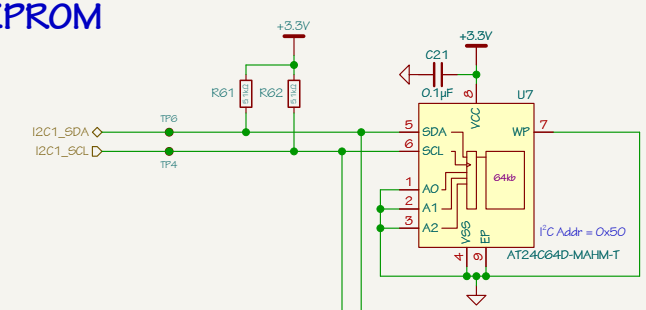
Size: A	Date: 2026-05-04	Rev: A
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Path /Microcontroller/Peripherals/

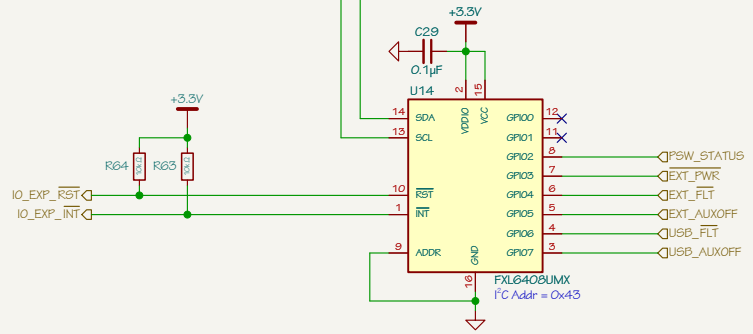
External ADC/GPIO



EEPROM

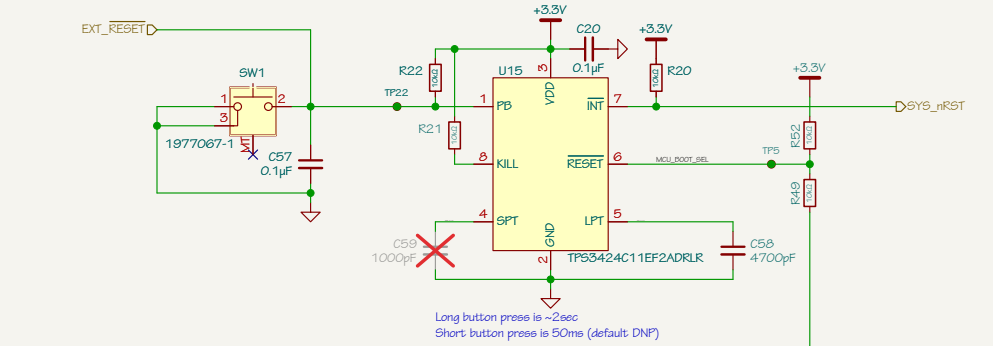


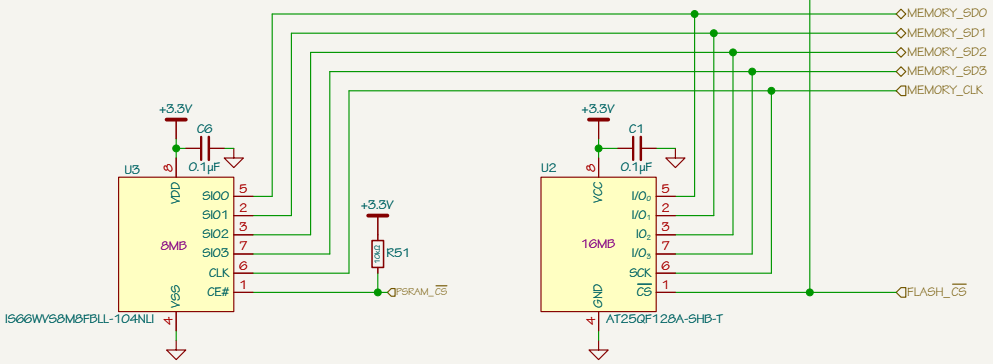
IO-EXPANDER



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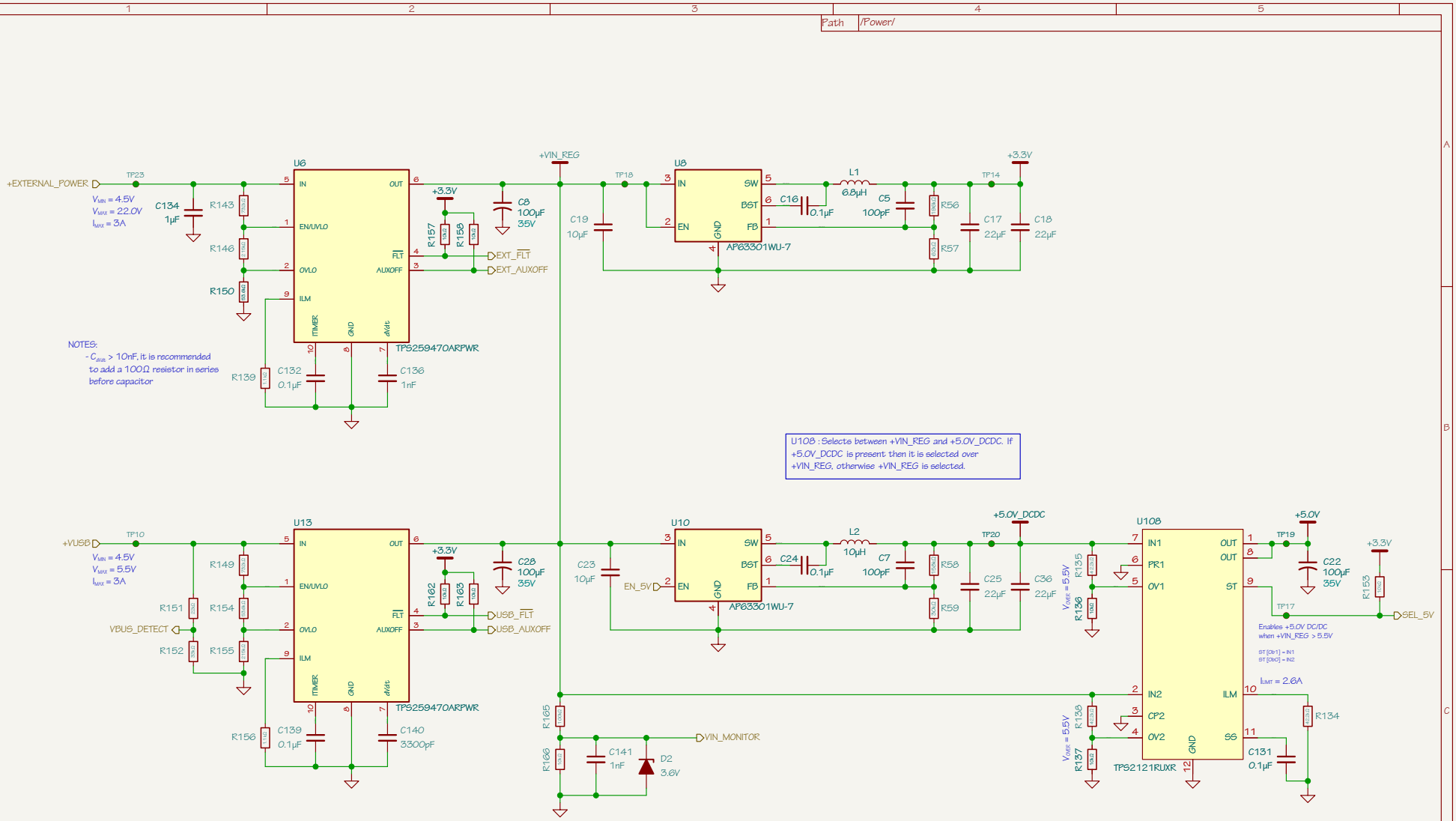
File: phyduino_pico_peripherals.kicad_sch		
Title: Phyduino Pico 2		
Size: A	Date: 2026-05-04	Rev: A
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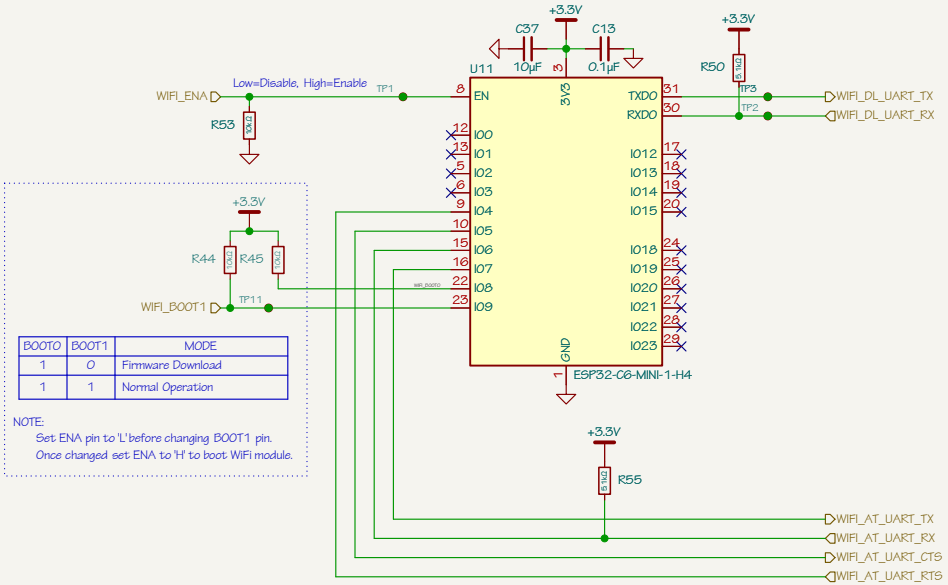
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File: phyduino_pico_memory.kicad_sch		
Title: Phyduino Pico 2		
Size: A	Date: 2026-05-04	**Rev: A**
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Path /Power/
+EXTERNAL_POWER
TP23
VMIN = 4.5V
VMAX = 22.0V
IMAX = 3A
C134 1µF
R143 100kΩ
R146 10kΩ
R150 95.1kΩ
U6
IN
OUT
EN/UVLO
OVLO
ILM
FLT
AUXOFF
ITIMER
GND
dV/dt
TPS259470ARPWR
+3.3V
R157 10kΩ
R158 10kΩ
C8 100µF 35V
EXT_FLT
EXT_AUXOFF
R139 1.1kΩ
C132 0.1µF
C136 1nF
NOTES:
- CdV/dt > 10nF, it is recommended to add a 100Ω resistor in series before capacitor
+VIN_REG
TP18
C19 10µF
U8
SW
BST
EN
FB
AP63301WU-7
L1 6.8µH
C16 0.1µF
C5 100pF
R56 200kΩ
R57 40.2kΩ
TP14
+3.3V
C17 22µF
C18 22µF
U108 : Selects between +VIN_REG and +5.0V_DCDC. If +5.0V_DCDC is present then it is selected over +VIN_REG, otherwise +VIN_REG is selected.
+VUSB
TP10
VMIN = 4.5V
VMAX = 5.5V
IMAX = 3A
R149 100kΩ
R151 30kΩ
R154 100kΩ
R152 30kΩ
R155 47.5kΩ
VBUS_DETECT
U13
TPS259470ARPWR
+3.3V
R162 10kΩ
R163 10kΩ
C28 100µF 35V
USB_FLT
USB_AUXOFF
R156 1.1kΩ
C139 0.1µF
C140 3300pF
C23 10µF
EN_5V
U10
AP63301WU-7
L2 10µH
C24 0.1µF
C7 100pF
R58 200kΩ
R59 40.2kΩ
TP20
+5.0V_DCDC
C25 22µF
C36 22µF
R135 100kΩ
VMIN = 5.5V
R136 10kΩ
U108
IN1
PR1
OV1
IN2
CP2
OV2
OUT
ST
ILM
SS
GND
TPS2121RUXR
TP19
+5.0V
C22 100µF 35V
TP17
Enables +5.0V DC/DC when +VIN_REG > 5.5V
ST [Ch1] = IN1
ST [Ch0] = IN2
ILMT = 2.6A
R153 10kΩ
SEL_5V
R133 100kΩ
R137 10kΩ
R134 82.5kΩ
C131 0.1µF
R165 100kΩ
R166 10kΩ
C141 1nF
D2 3.6V
VIN_MONITOR
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File: phyduino2_power_regs.kicad_sch
Title: Phyduino Pico 2
Size: A
Date: 2026-05-04
Rev: A
KiCad E.D.A. 10.0.4
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WiFi Module Interface

Path /Wireless Communication/



BOOT0	BOOT1	MODE
1	0	Firmware Download
1	1	Normal Operation

NOTE:
Set ENA pin to 'L' before changing BOOT1 pin.
Once changed set ENA to 'H' to boot WiFi module.

NOTES:

- To download ESP-AT firmware to ESP32 module use UART0 pins (TXD0 & RXD0).
- Both modules are pin compatiable, either can be used.
- Using ESP-AT as the network interface API.
- GitHub fork repo for ESP-AT: https://github.com/perc-sw/esp-at

ESP32-C61-MINI-1:

- ESP-AT version 5.0.1.0
- Hardware connection guide : https://docs.espressif.com/projects/esp-at/en/latest/esp32c61/Get_Started/Hardware_connection.html

Required Hardware connections

Connection	Pin	Direction
Download/Log RX	RXD0 (GPIO10)	INPUT
Download/Log TX	TXD0 (GPIO11)	OUTPUT
AT command/response RX	IO6 (GPIO6)	INPUT
AT command/response TX	IO7 (GPIO7)	OUTPUT
AT command/response CTS	IO5 (GPIO5)	OUTPUT
AT command/response RTS	IO4 (GPIO6)	INPUT

- 2.4GHz Wi-Fi 6 & BLE 5.0
- **Not for low power applications.**

ESP32-C6-MINI-1:

- ESP-AT version 4.1.1.0
- Hardware connection guide : https://docs.espressif.com/projects/esp-at/en/latest/esp32c6/Get_Started/Hardware_connection.html

Required Hardware connections

Connection	Pin	Direction
Download/Log RX	RXD0 (GPIO17)	INPUT
Download/Log TX	TXD0 (GPIO16)	OUTPUT
AT command/response RX	IO6 (GPIO6)	INPUT
AT command/response TX	IO7 (GPIO7)	OUTPUT
AT command/response CTS	IO5 (GPIO5)	OUTPUT
AT command/response RTS	IO4 (GPIO4)	INPUT

- 2.4GHz Wi-Fi 6 & BLE 5.0

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File: phyduino_pico2_wifi.kicad_sch

Title: Phyduino Pico 2

Size: A	Date: 2026-05-04	Rev: A
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